

Allegories & Algebra of Programming

cheatsheet — Freyd conventions; the B&dM laws in diagram order

§1. Conventions

Objects α, β, γ ; relations Q, R, S, T, U, V ; **maps** f, g, h, k, m ; coreflexives $C, D \subset 1$. RS = “first R , then S ” (diagram order); R° converse; $\subset$ containment; $\cap, \cup$ meet/join; $\emptyset$ empty; Π largest; 1 identity. $[\alpha]$ power-object, $\ni : [\alpha] \rightarrow \alpha$ membership (“epsilon-loff”), Λ power transpose, $\tau \triangleq \Lambda(1)$ singleton. F, G relators; μF initial algebra with structure map $\text{in} : F(\mu F) \rightarrow \mu F$; catamorphism $\langle S \rangle$. Points: $x R y$. **B&dM dictionary**: their $S \cdot R$ = our RS ; their $\in$ = our $\ni$ (their $\ni$ = our $\ni$); slashes flip keeping numerator/denominator roles: their $T \setminus U$ = our U/T , their U/T = our $T \setminus U$; their $\langle R| \rangle$ = our $\langle R| \rangle$; their α = our in .

§2. Adjunctions

Each row is a Galois connection $L \dashv U : LX \subset Y \iff X \subset UY$; $\cdot$ marks the argument slot. Every L preserves $\cup$, every U preserves $\cap$, both are monotone.

$\cdot \dashv \cdot$	$R^\circ \subset S \iff R \subset S^\circ$	(self-adjoint order-iso)
$\cdot \cap R \dashv R \Rightarrow \cdot$	$X \cap R \subset S \iff X \subset (R \Rightarrow S)$	
$T \cdot \dashv T \setminus \cdot$	$TX \subset U \iff X \subset T \setminus U$	
$\cdot T \dashv \cdot / T$	$XT \subset U \iff X \subset U/T$	
$\cdot - S \dashv S \cup \cdot$	$R - S \subset X \iff R \subset S \cup X$	
$f^\circ \dashv f \cdot$	shunting: $f^\circ R \subset S \iff R \subset fS$	
$\cdot f \dashv \cdot f^\circ$	shunting: $Rf \subset S \iff R \subset Sf^\circ$	
$\text{dom} \dashv \cdot \Pi$	$\text{dom } R \subset C \iff R \subset C\Pi$	(dually $\text{ran} \dashv \Pi \cdot$)
$\cdot^* \dashv \text{incl.}$	$R^* \subset X \iff R \subset X$ for preorders X	(reflection)
Λ iso	$f = \Lambda(R) \iff f \ni = R$: relations $\gamma \rightarrow \alpha \cong \text{maps } \gamma \rightarrow [\alpha]$	

§3. What preserves what

op	$\subset$	$\cap$	$\cup$	notes
$\cdot^\circ$	$=$	$=$	$=$	involution; $(RS)^\circ = S^\circ R^\circ$
$S \cdot$	mono	$\subset$; = if S simple	$=$	$S\emptyset = \emptyset$
$\cdot S$	mono	$\subset$; = if $SS^\circ \subset 1$	$=$	
relator F	mono	$\subset$; = on corefl.	$\supset$	$(FR)^\circ = F(R^\circ)$; maps $\mapsto$ maps; $F(\text{dom } R) = \text{dom}(FR)$
$T \setminus \cdot$	mono	$=$	$\supset$	antitone in T : $(R \cup S) \setminus T = (R \setminus T) \cap (S \setminus T)$
dom, ran	mono		$=$	
min	mono	$=$		$\text{min } R$ simple $\iff R$ antisymmetric
thin	mono			preorder if the argument is
μ	mono			$(\forall X. \varphi X \subset \psi X) \implies \mu\varphi \subset \mu\psi$

§4. Maps: products, sums, folds (B&dM ch. 2)

UP $\times$	$h = \langle f, g \rangle \iff h\pi = f \wedge h\pi' = g$
cancel, reflect	$\langle f, g \rangle \pi = f, \quad \langle f, g \rangle \pi' = g, \quad \langle \pi, \pi' \rangle = 1$
fuse	$m\langle h, k \rangle = \langle mh, mk \rangle$
functor	$f \times g \triangleq \langle \pi f, \pi' g \rangle, \quad (f \times g)(h \times k) = fh \times gk$
absorb	$\langle p, q \rangle (f \times g) = \langle pf, qg \rangle$
UP $+$	$h = [f, g] \iff \iota h = f \wedge \iota' h = g$
cancel, reflect	$\iota[f, g] = f, \quad \iota'[f, g] = g, \quad [\iota, \iota'] = 1$
fuse	$[h, k]m = [hm, km], \quad (h + k)[f, g] = [hf, kg]$
functor	$f + g \triangleq [f\iota, g\iota']$
exchange	$\langle [f, g], [h, k] \rangle = \langle [f, h], [g, k] \rangle$
UP $\langle \cdot \rangle$	$h = \langle f \rangle \iff \text{in } h = (Fh)f$
reflect	$\langle \text{in} \rangle = 1$
fuse	$f h = (Fh)g \implies \langle f \rangle h = \langle g \rangle$
shift	$\langle g f \rangle = \langle (Ff)g \rangle f$
Lambek	$\text{in iso}, \quad \text{in}^{-1} = \langle F \text{ in} \rangle$
type functor	$\text{T } f \triangleq \langle F(f, 1) \text{ in} \rangle$ (lists: map), $\text{in } (\text{T } f) = F(f, \text{T } f) \text{ in}$
map fusion	$(\text{T } g)\langle h \rangle = \langle F(g, 1) h \rangle$

§5. Program calculus (B&dM ch. 3)

banana-split	$\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle (F\pi)h, (F\pi')k \rangle \rangle$
mutual rec.	$\text{in } f = (F\langle f, g \rangle)h \wedge \text{in } g = (F\langle f, g \rangle)k \iff \langle f, g \rangle = \langle \langle h, k \rangle \rangle$
paramorphism triangle	$\text{in } f = (F\langle f, \langle h \rangle \rangle)g \implies f = \langle \langle g, (F\pi')h \rangle \rangle \pi$
Horner	$\text{tri } f \triangleq \langle F(1, \text{T } f) \text{ in} \rangle$
conditional	$gf = F(f, f)g \implies (\text{tri } f)\langle g \rangle = \langle F(1, f)g \rangle$
pre, post	$(p \rightarrow f, g) \triangleq p^? [f, g]; \quad (p \rightarrow f, f) = f$
filter	$(p \rightarrow f, g)h = (p \rightarrow fh, gh), \quad h(p \rightarrow f, g) = (hp \rightarrow hf, hg)$
UP curry	$(\text{T } f)(\text{filter } p) = (\text{filter}(fp))(\text{T } f)$
cancel, reflect	$g = \text{curry } f \iff (g \times 1) \text{ apply} = f$
fuse	$(\text{curry } f \times 1) \text{ apply} = f, \quad \text{curry}(\text{apply}) = 1$
	$g(\text{curry } f) = \text{curry}((g \times 1)f)$

structural recursion: if $\varphi(G(h \times 1)) = (Fh \times 1)\varphi$ then $(\text{in } \times 1)f = \varphi(Gf)h$ has the unique solution $f = (\langle \text{curry}(\varphi(G \text{ apply})h) \rangle \times 1) \text{ apply}$

§6. Allegory (B&dM 4.1–4.2)

converse	$(R^\circ)^\circ = R, \quad (RS)^\circ = S^\circ R^\circ, \quad (R \cap S)^\circ = R^\circ \cap S^\circ$
----------	--

modular law	$RS \cap T \subset (R \cap TS^\circ)S \subset (R \cap TS^\circ)(S \cap R^\circ T)$
modular identity	$R \cap TS = R \cap (T \cap RS^\circ)S$
	$R \subset RR^\circ R; \quad S \text{ simple} \implies S = SS^\circ S$
indirect equality	$R = S \iff (\forall X. X \subset R \iff X \subset S)$
dom, ran	$\text{dom } R \triangleq 1 \cap RR^\circ, \quad \text{ran } R \triangleq 1 \cap R^\circ R$
restrict	$R = (\text{dom } R)R = R(\text{ran } R)$
ran laws	$\text{ran}(SR) = \text{ran}((\text{ran } S)R), \quad \text{ran}(R \cap S) = 1 \cap S^\circ R$
	$\text{dom } S = 1 \cap S\Pi, \quad \text{ran}(RC) = C \cap \text{ran } R$
coreflexives	$CD = C \cap D = DC, \quad (RC) \cap S = (R \cap S)C$
entire	$1 \subset RR^\circ; \quad R \cap S \text{ entire} \iff 1 \subset RS^\circ$
simple	$R^\circ R \subset 1; \quad \text{map} = \text{entire} + \text{simple}, \quad f \subset g \implies f = g$
simple dist.	$S \text{ simple: } RS \cap T = (R \cap TS^\circ)S, \quad S(R \cap T) = SR \cap ST$
converse-free maps	$SR \subset 1 \wedge 1 \subset RS \implies S = R^\circ$ and R a map
orders	$\text{sym} \wedge \text{trans} \iff R = R^\circ R; \quad \text{preorder} \iff R = R/R \iff R = fSf^\circ, S \text{ an order}$

§7. Tabulations, unit (B&dM 4.3)

tabulation	maps f, g with $f^\circ g = R, \quad ff^\circ \cap gg^\circ = 1$ (jointly monic)
factorisation	$h^\circ k \subset R \iff \exists! \text{ map } m: h = mf \wedge k = mg$
monic, cover	$mm^\circ = 1; \quad c^\circ c = 1; \quad \text{every map } f = cm$
legs	$R \text{ simple} \implies f \text{ monic}; \quad R \text{ entire} \implies f \text{ cover}; \quad R \text{ a map} \implies f \text{ iso}$
unit	$R : U \rightarrow U \implies R \subset 1$; entire map $!_\alpha : \alpha \rightarrow U; \quad \Pi \triangleq !^!$

Horn meta-theorem: a Horn sentence over $\cap, ^\circ$, composition, tabulations, unit holds in all unitary tabular allegories $\iff$ it holds in Rel

§8. Division, negation (B&dM 4.4–4.5)

pointwise	$x(T \setminus U)y \iff \forall w. wT \Rightarrow wUy; \quad / \text{ dual}$
cancel	$T(T \setminus U) \subset U, \quad (U/T)T \subset U$
associate	$(TU) \setminus V = U \setminus (T \setminus V), \quad V/(UT) = (V/T)/U$
maps	$f(S/R) = (fS)/R, \quad (S/R)f^\circ = S/(fR); \quad ^\circ\text{-duals for } \setminus$
symmetric div.	$(R/S) \cap (S/R)^\circ; \quad \Lambda(S)\Lambda(R)^\circ = (S/R) \cap (R/S)^\circ$
negation	$\neg R \triangleq (R \Rightarrow \emptyset); \quad \neg(R \cup S) = \neg R \cap \neg S$
boolean	$\iff \neg \neg R = R \iff R \cup \neg R = \Pi$ (Rel is boolean)
Schröder	$SR \subset T \iff S^\circ(\neg T) \subset \neg R \iff (\neg T)R^\circ \subset \neg S$
neg. division	$T \setminus (\neg U) = \neg(T^\circ U)$ (always); boolean: $T \setminus U = \neg(T^\circ(\neg U))$
lexical	$R; S \triangleq R \cap (R^\circ \Rightarrow S)$ — preorders closed, unit Π
strict part	$ R \triangleq R \cap \neg R^\circ$
Galois, point-free	orders as $R, S: f(x) \preceq y \iff x \preceq g(y)$ is $Rf^\circ = gS$

§9. Power (B&dM 4.6)

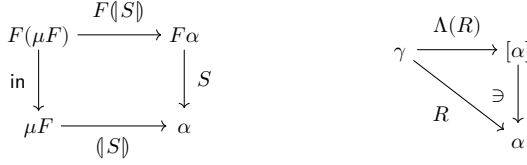
cancel	$\Lambda(R)\ni = R; \quad \Lambda(R)$ is the unique map with this property
fuse, reflect	$\Lambda(fR) = f\Lambda(R), \quad \Lambda(\ni) = 1$
tabular	$\Lambda(R) = (R/\ni) \cap (\ni/R)^\circ$
existential image	$ER \triangleq \Lambda(\ni R)$ (Freyd’s covariant $[\cdot]$, §1.922); $E1 = 1$
absorb	$\Lambda(S)(ER) = \Lambda(SR); \quad \text{hence } (ET)(ER) = E(TR)$
naturality	$(ER)\ni = \ni R$
power functor	$P \triangleq E$ on maps; τ monic, $\Lambda(f) = f\tau, \quad \tau(Pf) = f\tau$
monad	$\bigcup \triangleq E\exists : [[\alpha]] \rightarrow [\alpha]; \quad \tau \bigcup = 1 = (P\tau) \bigcup, \quad \bigcup \bigcup = (P \bigcup) \bigcup$
decompose	$\Lambda(R) = \tau(ER), \quad ER = (PAR) \bigcup$
$\ni$ -cancel	$(\ni/R) \setminus \ni = R$
inclusion	$\ni/\ni : [\alpha] \rightarrow [\alpha], x \mapsto \text{subsets of } x$; preorder, antisymmetric iff extensional

§10. Datatypes in allegories (B&dM ch. 5)

relator	monotone functor $\iff$ preserves $^\circ$; relators agreeing on maps are equal
pairing	$\langle R, S \rangle \triangleq R\pi^\circ \cap S\pi^\circ, \quad R \times S \triangleq \langle \pi R, \pi' S \rangle$
absorb	$\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle; \quad f\langle R, S \rangle = \langle fR, fS \rangle$ (f a map)
projections	$\langle R, S \rangle \pi = (\text{dom } S)R, \quad \langle R, S \rangle \pi' = (\text{dom } R)S$ — $\times$ is not a product
cancel	$\langle X, Y \rangle \langle R, S \rangle^\circ = XR^\circ \cap YS^\circ$
junc	$[R, S] \triangleq \iota^\circ R \cup \iota^\circ S = [\Lambda R, \Lambda S] \ni$ — a genuine coproduct
cancel	$\iota[R, S] = R, \quad [U, V]^\circ[R, S] = U^\circ R \cup V^\circ S$
functors	$(R \times S)(U \times V) = RU \times SV, \quad (R + S)(U + V) = RU + SV$
guard	$\sim C: C \cap \sim C = \emptyset, C \cup \sim C = 1; \quad \text{guard } C \triangleq [C, \sim C]^\circ$ a map
conditional	$(C \rightarrow R, S) \triangleq (\text{guard } C)(R + S)$
UP	$R \subset (C \rightarrow S, T) \iff CR \subset S \wedge (\sim C)R \subset T$
power relator	$PR \triangleq ((\ni R)/\ni) \cap (\ni \setminus (R\exists))$; on maps $P = E$
lax naturality	$\varphi : F \hookrightarrow G \triangleq \varphi(R) \supset (GR)\varphi; \quad \iff$ natural on maps. E.g. $(PR)\ni \subset \ni R$

Eilenberg–Wright	$\text{in } X = (FX)S \iff X = \llbracket S \rrbracket; \quad \Lambda \llbracket S \rrbracket = \llbracket \Lambda((F\exists)S) \rrbracket$
type relators	$\text{in } (\top R) = F(R, \top R) \text{ in}, \quad (\top R)^\circ = \top(R^\circ); \llbracket R \rrbracket \text{ entire/a map if } R \text{ is}$
transposes	$\text{cp}(F) \triangleq \Lambda(F\exists); \quad \Lambda(R \cup S) = \langle \Lambda R, \Lambda S \rangle \text{ cup, same for } \cap / \text{“cap”}, \times / \text{“cross”}$
vocabulary	$\text{subseq} = \llbracket [\text{nil}, \text{cons} \cup \pi'] \rrbracket, \quad \text{prefix} = \text{cat}^\circ \pi, \quad \text{perm} = \text{bagify bagify}^\circ, \text{partition} = \text{concat}^\circ$

§11. Recursion, fixed points, closure (B&dM ch. 6)



Knaster–Tarski	φ monotone: least solutions of $\varphi X \subset X$ and $\varphi X = X$ coincide ($\mu\varphi$); greatest dually
computation	$\varphi(\mu\varphi) = \mu\varphi; \quad \mu\varphi = \bigcap \{X : \varphi X \subset X\}$
induction	$\varphi X \subset X \implies \mu\varphi \subset X; \quad \varphi$ continuous: $\mu\varphi = \bigcup_n \varphi^n \emptyset$
cata	$\llbracket R \rrbracket = \text{the unique } X \text{ with } X = \text{in}^\circ(FX)R$
least, greatest	$\text{in}^\circ(FX)R \subset X \implies \llbracket R \rrbracket \subset X$, and dually
fuse	$(FS)T \subset RS \implies \llbracket T \rrbracket \subset \llbracket R \rrbracket S$
fuse (dual)	$RS \subset (FS)T \implies \llbracket R \rrbracket S \subset \llbracket T \rrbracket$
compare	$(F\langle S \rangle)R \subset (F\langle S \rangle)S \implies \llbracket R \rrbracket \subset \llbracket S \rrbracket$
rolling	$\mu(X \mapsto \varphi(\psi X)) = \varphi(\mu(X \mapsto \psi(\varphi X)))$
diagonal	$\mu(X \mapsto \mu(Y \mapsto \varphi(X, Y))) = \mu(X \mapsto \varphi(X, X))$
inductive	R inductive $\triangleq \forall X. X/R \subset X \implies \Pi \subset X$
	S inductive and $(R \subset S \text{ or } RR \subset SR) \implies R$ inductive; $S \iff S^+$; meets too
well-founded	$\forall X. X \subset RX \implies X = \emptyset$; inductive $\implies$ wf, conversely in boolean; fRf° wf if R wf
membership	$\text{mem } \text{Id} = 1, \text{mem } K_\beta = \emptyset, \text{mem}(F + G) = [\text{mem } F, \text{mem } G]$
	$\text{mem}(F \times G) = \pi(\text{mem } F) \cup \pi'(\text{mem } G), \quad \text{for } F(G \cdot): (\text{mem } F)(\text{mem } G), \text{mem } [\cdot] = \exists$
largest lax	$(FR)(\text{mem } F) \subset (\text{mem } F)R; \quad \text{in}^\circ(\text{mem } F)$ inductive (immediate subterm)
unique fixpoint	$S(\text{mem } F)$ inductive $\implies X = S(FX)R$ has a unique solution; entire if R, S entire
functional	$g(\text{mem } F)$ inductive $\implies$ the unique solution of $X = g(FX)f$ is a map
converse cata	$\text{ran } R = 1 \text{ and } Rf \subset (Ff) \text{ in} \implies f^\circ = \llbracket R \rrbracket$
closure	$R^* = \mu(X \mapsto 1 \cup RX) = \mu(X \mapsto 1 \cup XR); \quad \text{unique fixed point iff } R \text{ inductive}$
	$SR^* = \mu(X \mapsto S \cup XR), \quad (R \cup S)^* = (R^*S)^*R^*, \quad R^* = \llbracket [1, 1] \rrbracket \circ \llbracket [1, R] \rrbracket$
sorting	$\text{sort} \subset \text{perm}(\text{ordered } R), R$ a connected preorder ($R \cup R^\circ = \Pi$)
	$(\text{ordered } R)(\text{ordered } S) = \text{ordered}(R \cap S); \text{ sorts emerge by cata-fusion + hylo}$

Theorem (Hylomorphism).

$$\llbracket S \rrbracket \circ \llbracket R \rrbracket = \mu(X \mapsto S^\circ(FX)R)$$

Divide (S°), conquer (FX), combine (R); the virtual data structure is μF . Coproduct split ($F = G + H$): $\mu(X \mapsto S_1^\circ(GX)R_1 \cup S_2^\circ(HX)R_2)$. Hylos of simple parts are simple.

§12. Optimisation: min & max (B&dM 7.1)

definition	$\min R \triangleq \exists \cap (\exists^\circ \setminus R) : [\alpha] \rightarrow \alpha, \quad \max R \triangleq \min R^\circ$
UP	$X \subset \min R \iff X \subset \exists \wedge \exists^\circ X \subset R$
divide by $\exists$	$\tau(\exists^\circ \setminus R) = R, \quad \Lambda(S)(\exists^\circ \setminus R) = S^\circ \setminus R, \quad \bigcup(\exists^\circ \setminus R) = \exists^\circ \setminus (\exists^\circ \setminus R)$
singleton	$\tau(\min R) = 1 \cap R$
transpose	$\Lambda(S)(\min R) = S \cap (S^\circ \setminus R) \quad (= \text{AoPA shrink } S \upharpoonright R)$
UP	$X \subset \Lambda(S)(\min R) \iff X \subset S \wedge S^\circ X \subset R$
context	$\Lambda(S)(\min R) = \Lambda(S)(\min(R \cap S^\circ S))$
image	$(ES)(\min R) = \exists S \cap ((\exists S^\circ) \setminus R)$
maps	$(Pf)(\min R) = (\min(fRf^\circ))f$
relations	$(PS)(\min R) \subset \exists S \cap (\exists^\circ \setminus (SR)), \text{ if } R \text{ reflexive}$
union	R preorder: $(P(\min R))(\min R) \subset \bigcup(\min R) = (P(\text{dom } \min R)) \bigcup (\min R)$
meet	$\min(R \cap S) = \min R \cap \min S$
reflexive	$R = \exists^\circ(\min R); \quad \min R \subset \min S \iff R \subset S$ (S reflexive too)
preorder	$\min R = \exists \cap (\min R)R, \quad \Lambda(R)(\max R) = R \cap R^\circ$
pairs	$\langle S(\min R), T(\min R) \rangle \subset \Lambda(S, T)(\min(R \times R))$
lexical	R reflexive, S preorder: $\Lambda(\min S)(\min R) = \min(S; R)$

minimal elts.	$\text{mnl } R \triangleq \min(R^\circ \Rightarrow R); \quad \min R \subset \text{mnl } R, = \text{ iff } R \text{ connected preorder}$
well-bounded	$\text{dom } \exists = \text{dom}(\min R) \iff R \text{ wf} \iff \exists \subset (\min R)R^\circ$

§13. Greedy (B&dM 7.2)

monotonic	S monotonic on $R \triangleq (FR)S \subset SR$
for maps	$\iff f^\circ(FR)f \subset R \iff FR \subset fRf^\circ; \quad \text{on } R \iff \text{on } R^\circ$ (maps only)
distributivity	f monotonic on $R \iff (F(\min R))f \subset \Lambda((F\exists)f)(\min R)$

Theorem (Greedy).

R a preorder, S monotonic on R° :

$$\llbracket \Lambda(S)(\min R) \rrbracket \subset \Lambda(\llbracket S \rrbracket)(\min R)$$

Max form (S monotonic on R): $\llbracket \Lambda(S)(\max R) \rrbracket \subset \Lambda(\llbracket S \rrbracket)(\max R)$. Context: monotonicity on $R^\circ \cap \llbracket S \rrbracket \llbracket S \rrbracket^\circ$ suffices. Variant: f monotonic on $R, f \subset \Lambda(S)(\min R) \implies \llbracket f \rrbracket \subset \Lambda(\llbracket S \rrbracket)(\min R)$.

optimal substeps: S monotonic on $R^\circ \implies (\min(FR))\Lambda(S)(\min R) \subset (ES)(\min R)$

§14. Thinning (B&dM 8.1, 8.3)

definition	$\text{thin } Q \triangleq (\exists/\exists) \cap (\exists^\circ \setminus (Q\exists^\circ)) : [\alpha] \rightarrow [\alpha]$
	shrink a set, keeping a Q -bound for each dropped element
UP	$X \subset \Lambda(S)(\text{thin } Q) \iff X \exists \subset S \wedge S^\circ X \subset Q\exists^\circ$
order	$1 \subset \text{thin } Q; \quad \text{thin } 1 = 1$
introduction	$Q \subset R$ (preorders) $\implies \min R = (\text{thin } Q)(\min R)$
elimination	$(\min Q)\tau \subset \text{thin } Q; \quad R \cap S^\circ S \subset Q \implies \Lambda(S)(\min R)\tau \subset \Lambda(S)(\text{thin } Q)$
singleton	$\min R = (\text{thin } R)\tau^\circ$
context	$\Lambda(S)(\text{thin } Q) = \Lambda(S)(\text{thin}(Q \cap S^\circ S))$
union	$(P(\text{thin } Q)) \bigcup \subset \bigcup(\text{thin } Q)$
minimal elts.	Q well-supported $\implies \Lambda(\text{mnl } Q) \subset \text{thin } Q$

Theorem (Thinning).

Q a preorder, S monotonic on Q° :

$$\llbracket \Lambda((F\exists)S)(\text{thin } Q) \rrbracket \subset \Lambda(\llbracket S \rrbracket)(\text{thin } Q)$$

With $Q \subset R$, postcompose $\min R$: $\llbracket \Lambda((F\exists)S)(\text{thin } Q) \rrbracket(\min R) \subset \Lambda(\llbracket S \rrbracket)(\min R)$ — keep a Pareto front of partial solutions, pick the optimum at the end.

sorted lists	$\text{sort } P \triangleq \text{setify}^\circ(\text{ordered } P); \quad (\text{sort } P)(\text{thinlist } Q) \subset (\text{thin } Q)(\text{sort } P)$
merge	$((\text{sort } P) \times (\text{sort } P))(\text{merge } P) \subset \text{cup}(\text{sort } P)$
connected Q	$\text{thinlist } Qx = [\text{minlist } Qx]$

binary thinning: $S = f_1 p_1 \cup f_2 p_2$ (p_i coreflexive, $f_i p_i$ monotonic on Q° , f_i monotonic on a connected preorder $P, Q \subset R$): fold “apply both steps to sorted lists, merge by P , bump-thin by Q ”, finish with $\text{minlist } R$ — refines $\Lambda(\llbracket S \rrbracket)(\min R)$ on lists

§15. Dynamic programming (B&dM 9.1)

Theorem (Dynamic programming).

h a map monotonic on preorder $R; \quad \mathcal{H} \triangleq \llbracket T \rrbracket^\circ \llbracket h \rrbracket$:

$$\mu(X \mapsto \Lambda(T^\circ)P((FX)h)(\min R)) \subset \Lambda(\mathcal{H})(\min R)$$

With thinning: insert $\text{thin } Q$ after $\Lambda(T^\circ)$, provided Q is a preorder with $Q^\circ(F\mathcal{H})h \subset (F\mathcal{H})hR^\circ$. Decompose all ways, solve subproblems, thin, keep an optimum.

uniqueness	the fixed point is unique if $T^\circ(\text{mem } F)$ is inductive; entire (an implementable recursion) if $\Lambda(T^\circ)$ yields finite non-empty sets and R is a connected preorder
cost	$R = \text{cost leq cost}^\circ, \quad h \text{ cost} = (F \text{ cost})k, \quad (F \text{ leq})k \subset k \text{ leq} \implies h$ monotonic on R
finding Q	preorders U, V with $F(U, R)h \subset hR$ and $V^\circ \mathcal{H} \subset \mathcal{H}R^\circ \implies Q = F(U, V)$ works (often $U = \Pi$)
greedy from DP	bifunctor conditions collapse the DP recursion to a greedy sweep (B&dM ch. 10)